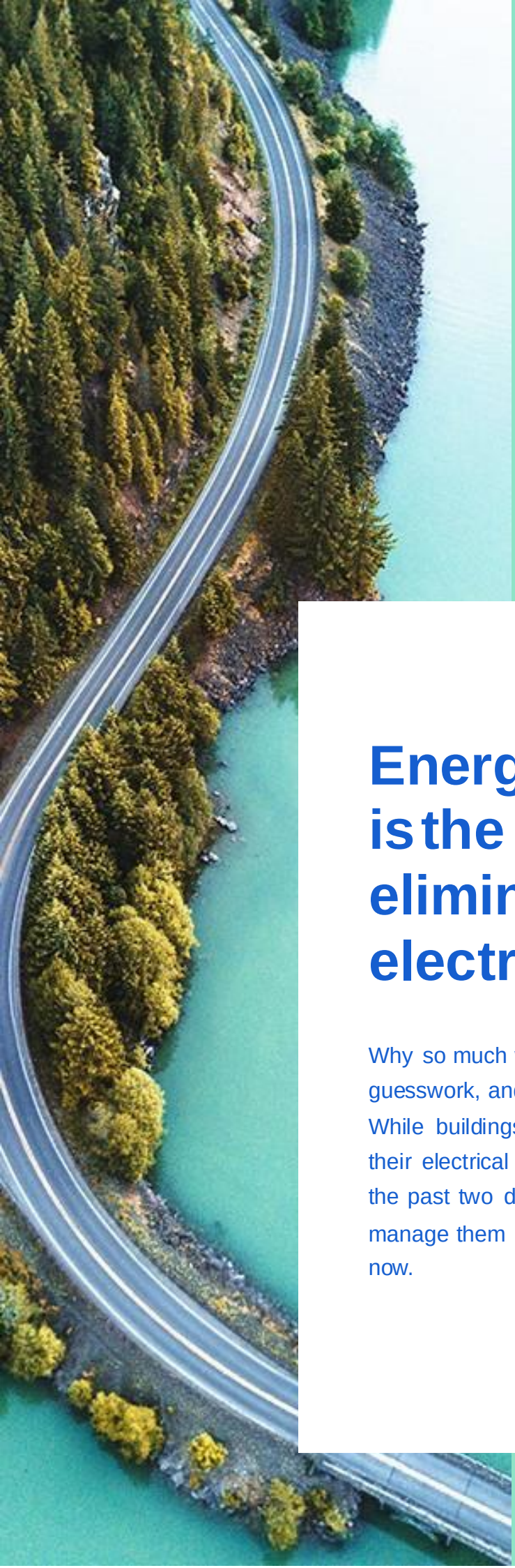
Amp

Transforming your built environments to Intelligent & Efficient spaces with Energy Analytics

IOT & Analytics Solution

V2.0



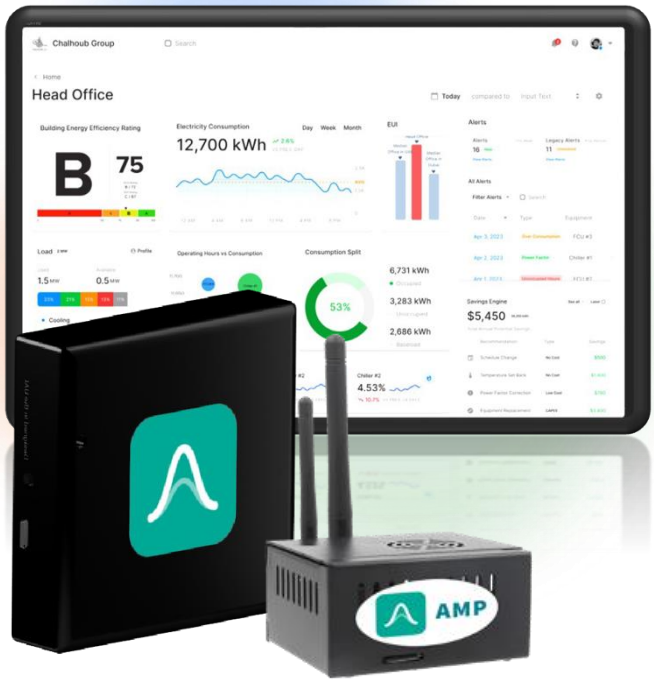
Energy Intelligence is the key to eliminating electricity waste

Why so much waste? Because buildings run on guesswork, and energy analysis is hard.
While buildings themselves and the complexity of their electrical load have significantly changed over the past two decades, the tools and processes to manage them haven't evolved to keep up - until now.

Our Integrated Solution

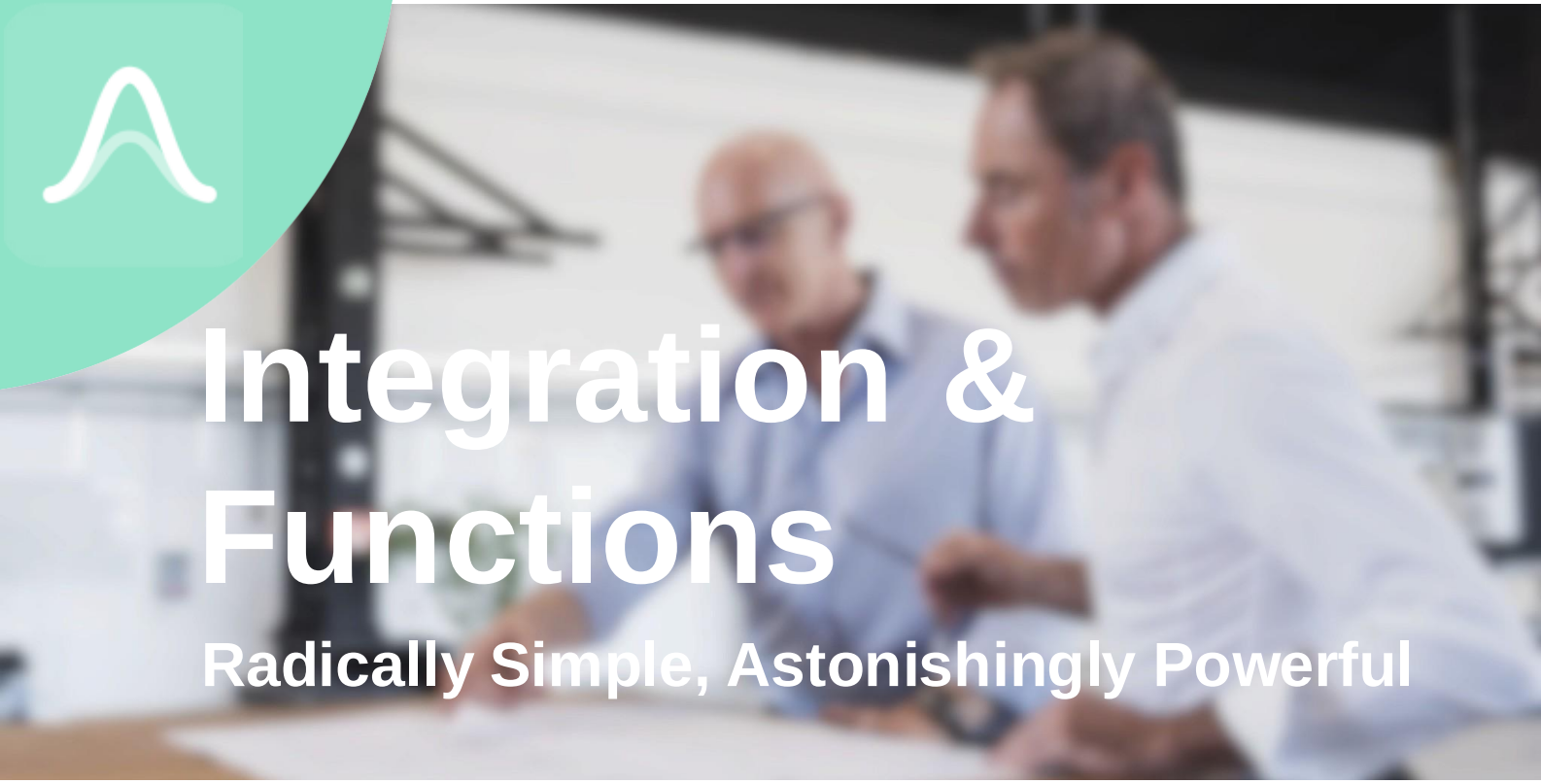
Product Overview

AmpEnterprise V1.2
Energy Platform V2.0



Providing unmatched connectivity with seemless integration and interface

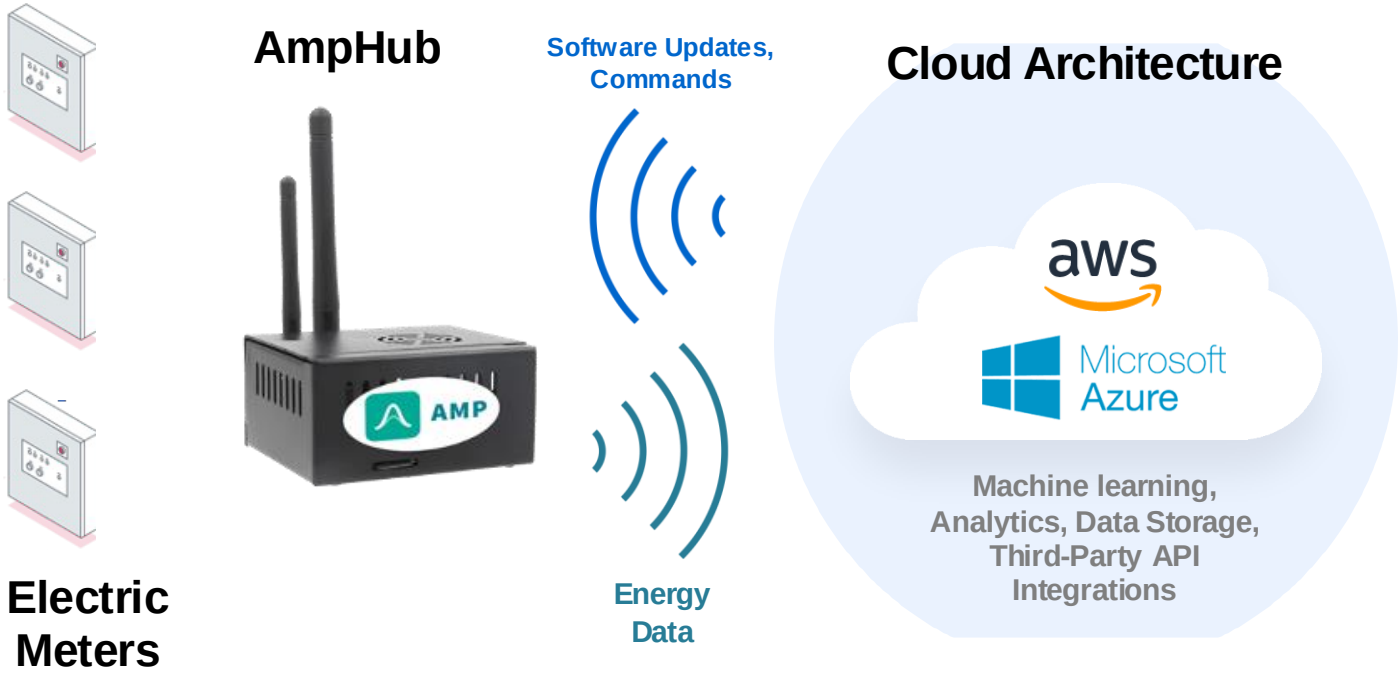
- Fully integrated solution including *all* required hardware and software configuration
- Amp Hub functions as a gateway for all devices on premises.
- Web-connected platform: provides access to users anywhere, at all times
- Powerful IOT device combining functions of an edge device and gateway
- Real-time data analytics and alerts: data sampling at 1-min intervals provides instantenous feedback to users
- Scaleable architecture connecting at the device level for equipment data and analytics



Integration & Functions

Radically Simple, Astonishingly Powerful

Functionality



- Plug and Play installation by Amp certified partners
- Default data-sampling at 1-minute interval.
- Amp Hubs are connected to any wall socket or plug.
- Aggregate equipment data at the panel level. Unlimited panel coverage in small/medium/large buildings.
- Transmit data using local encrypted protocol to the Hub
- No legacy equipment, network skills or third-party sensors required to be purchased or exercised by the customer.
- All communications and data is encrypted and processed to ensure safety.

Integration

Basic Function

- Multi-meter Gateway for commercial & industrial purposes
- Analyzes kWh, current, voltage, power, reactive energy, reactive power, power factor, frequency
- LED indicates connectivity
- Size: 70x57x35mm

Accuracy	Class 0.5
Range	~150m
Communications	WiFi, LAN, GSM, Bluetooth, ZigBee, NB-IOT
Weight	165g
Sample Rate	1minute: 15min aggregate

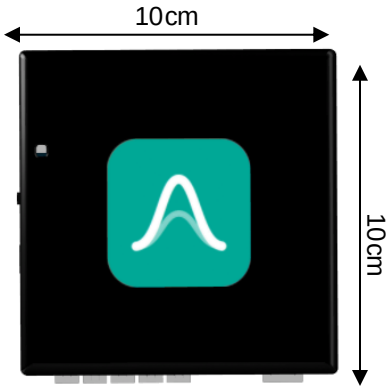
IOT Solution

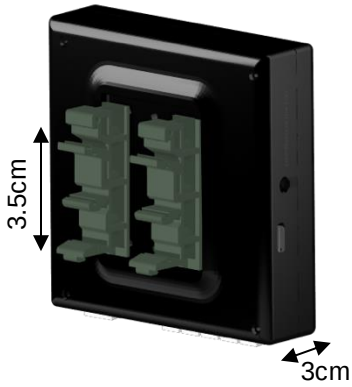
Basic Function

- Suitable for up to 50A current
- Applicable voltage between 120V – 240V
- Measures kWh, current, voltage, power, reactive energy, reactive power, power factor, frequency
- LED indicates connectivity
- Small size: 10x10x3cm
- Communications include WiFi, Bluetooth, GSM
- Din-rail mounting

Accuracy	Class 0.5
Power Consumption	< 5W
Frequency	50Hz/60Hz
Weight	200g
CT Ports	12
Sample Rate	8,000 samples/sec

Dimension

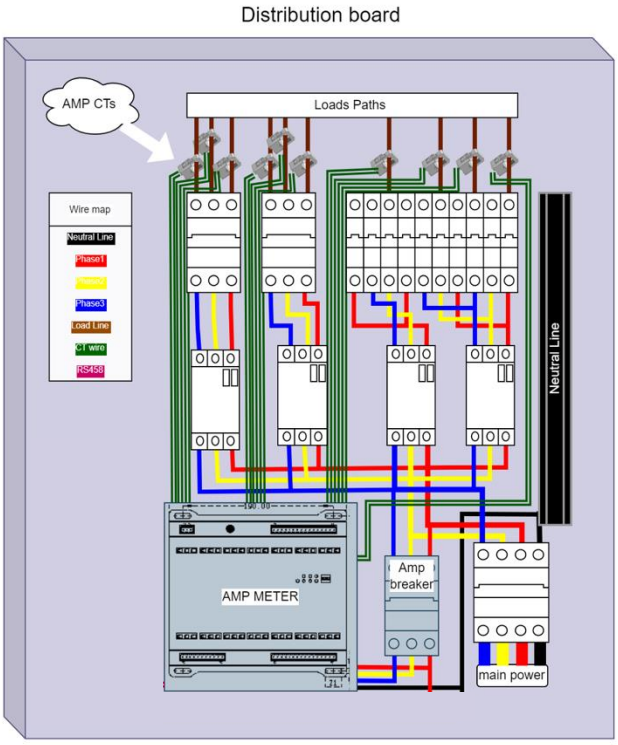




Integration

Detail of installation

- **The solution is mounted onto the selected electrical panels din rail. If no space is available, it is possible to mount against the wall of the distribution board.**
- **The solution uses energy sensors (CTs) that sample equipment energy data and feed into the Energy Meter. These are compatible with both single phase and three phase devices**
- **All data is transmitted to the Amp Hub in real-time, where further processing and analytics takes place. All data is then sent to the cloud.**
- **An onboarding link will be emailed to the customer or integrations team to connect the Hub online after successfully integrating the solution.**



Products used	
Product	Function
Amp Hub	Gateway & Data Analyzer
CT Sensors	Data Samplers
Power Cables	Powering Kit
Energy Meter	Data Collection

Details of Architecture

Hardware Considerations

Amp conforms to the general commercial standards of the relevant industry or the required standard of Dubai Municipality as referenced in the *Dubai Building Code 2021*, the relevant sections of which are represented below. Furthermore, a warranty shall be extended to all devices for one year – and they have an expected life expectancy that exceeds five years.

No.	Description		Specification
1	Type of meters		Static type kWh meter
2	Reference voltages and frequency		See G.4.2
3	**Accuracy class** Meter application	System rating	
3.1	Direct metering on LV system	≤120 A	Class 1
3.2	CT metering on LV system	>125 A	0.5s
4	**Register**		
4.1	Type of register		LCD
4.2	Number of digits		8 (minimum)
4.3	Height of numerals		8 mm (minimum)
4.4	LCD screen size		80 mm × 20 mm or approved by DEWA
5	**Service conditions**		
5.1	Operating temperature range (minimum)		700 °C
5.2	Relative humidity		100%
5.3	Transport and storage temperature range (minimum)		850 °C
6	Type of communication		Modular type with detachable communication module (to choose wired or wireless communication system)
7	Meter communication ports		Optical. M-bus, RS485, etc.

Table G.1 Electricity meter specifications

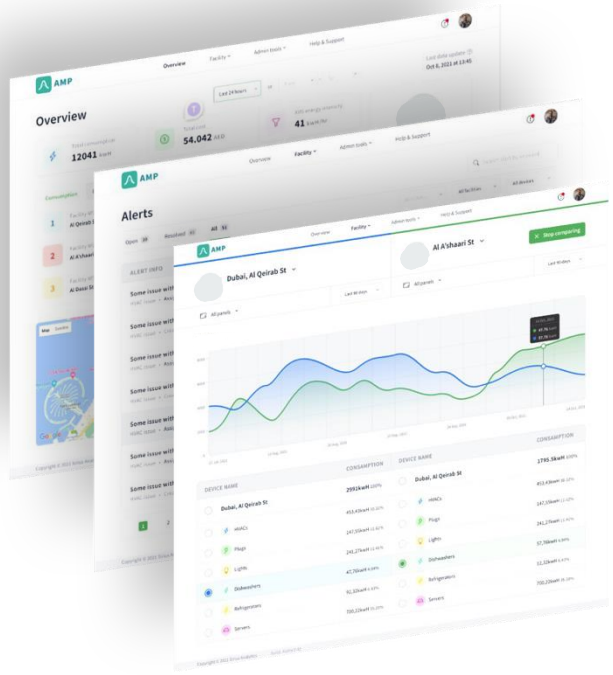
No.	Description	Specification
1	Rated secondary current	5 A
2	Maximum primary voltage	600 V
3	Rated frequency	50 Hz
4	Rated primary current (only one is applicable)	200 A, 300 A, 400 A, 800 A, 1,600 A or 2,400 A
5	Rated burden	5 VA
6	Number of phases	1
7	Accuracy class CT application CT metering on LV system	
8	Medium of installation	Air
9	Size of busbars (primary) **Rated primary current** a) 200 A, 300 A b) 400 A c) 600 A, 800 A d) 1,200 A, 1,600 A e) 2,400 A	 **Busbar size** 20 mm × 10 mm 30 mm × 10 mm 50 mm × 10 mm or 2 number 30 mm × 10 mm 2 number 60 mm × 10 mm 2 number 80 mm × 10 mm

Table G.2 Current transformers (CT) specifications

The Platform

Radically Simple, Astonishingly Powerful

Energy Platform V1.1

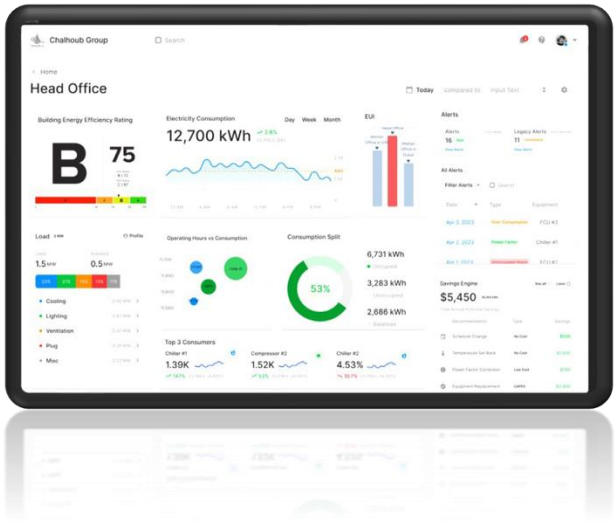


Unparalelled data and insights at your fingertips

- Centralized Portfolio, Building and Equipment level data provided in real-time and unlimited historical access
- Web-based platform providing access anytime, anyplace around the world
- Sleek and intuitive interface with captivating data visualization to drive real data-driven decisions
- Unlimited users with rules-based access and credentials
- Alarms, graphs, anomaly detection and trend setting features
- Compare equipment and buildings across your portfolio
- Data available online within 24 hours of IOT integration

Transparent Software License

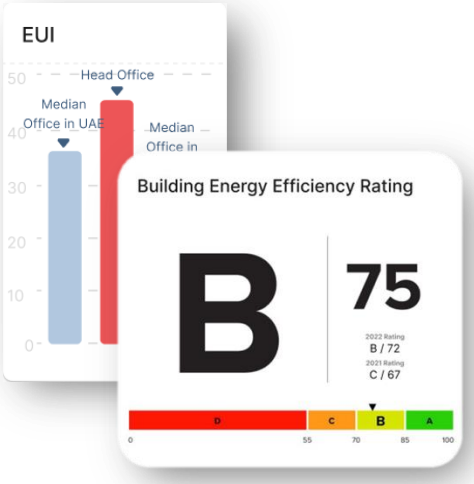
- Subscription terms vary from monthly, annually or multi-year
- Price per device per subscription term
- Billing term as per the subscription term
- Continuous upgrades and software updates rolled-out throughout subscription term duration
- All required connected devices included within integration or subscription fees
- Third party data-point customization available



Key Features

Unlock Insights, Drive Change

Energy Platform V1.1

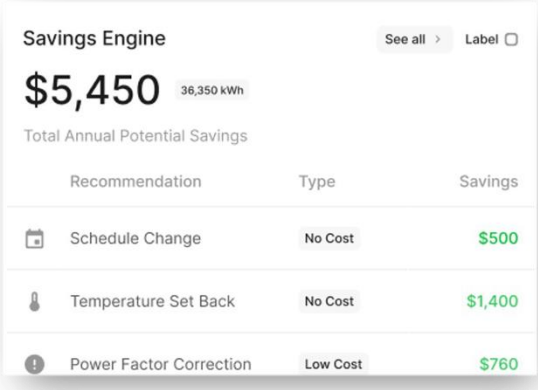


Monitor and Report Efficiency

Empower your team and increase productivity by benchmarking energy efficiency and performance. Set targets for portfolio assets and identify leaders and laggers. Utilize industry leading EUI benchmarks to track performance, and compare assets.

Identify Inefficiencies & Savings

Find and verify energy savings projects without manual analysis. You shouldn't have to be a data expert to reach your net-zero or operational excellence goals. We identify anomalous behavior and recommend opportunities to improve your efficiency and reduce your footprint



Key Features

Unlock Insights, Drive Change

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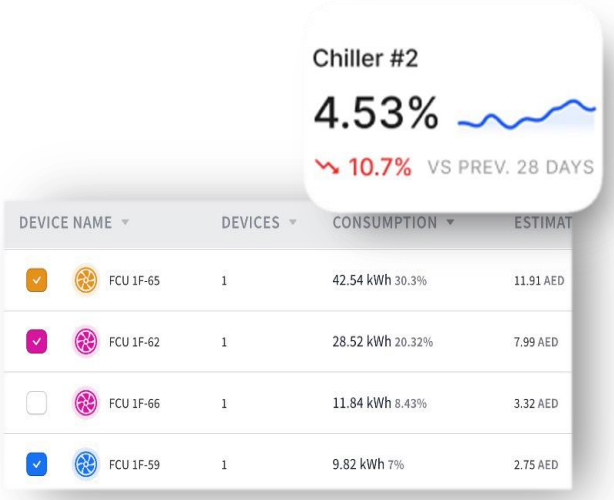
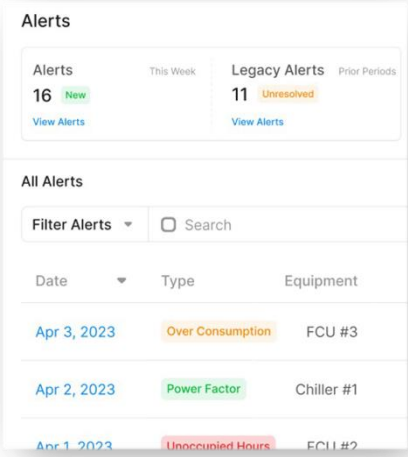


Capture Data from every Asset

Amp automates data collection down to every device and equipment within a portfolio. Get rid of guesswork and implement data-driven decision making. With up to 10 data points per device, our unparalleled analytics magnifies data and uncovers performance like never before.

Alerts Notifications

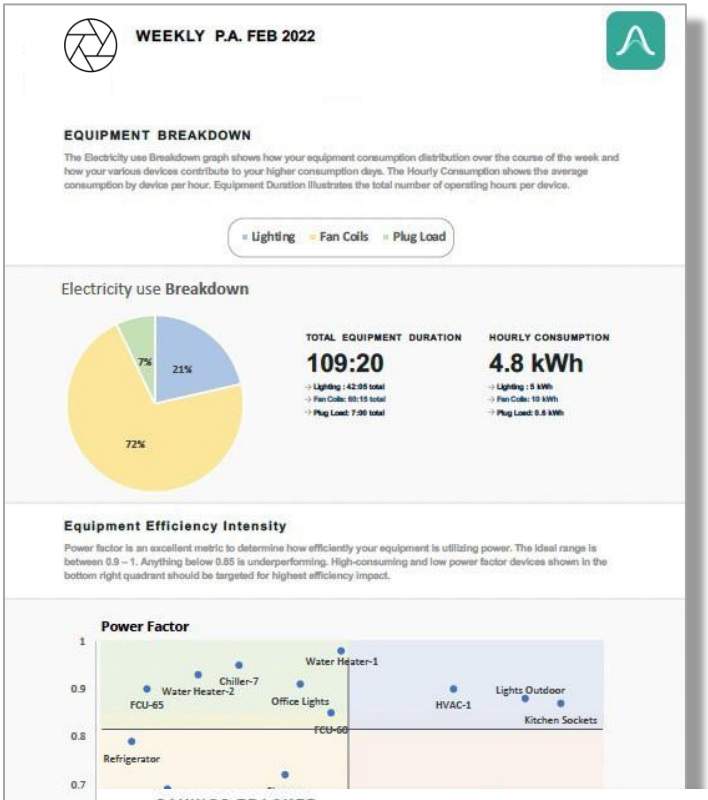
Receive real-time and big data driven alerts and anomaly detection via e-mail, sms or the platform. Parameters built by analytics engine based on baseline performance, universal technical standards and best practices, and may include user preferences. Quantify cost and drive action.

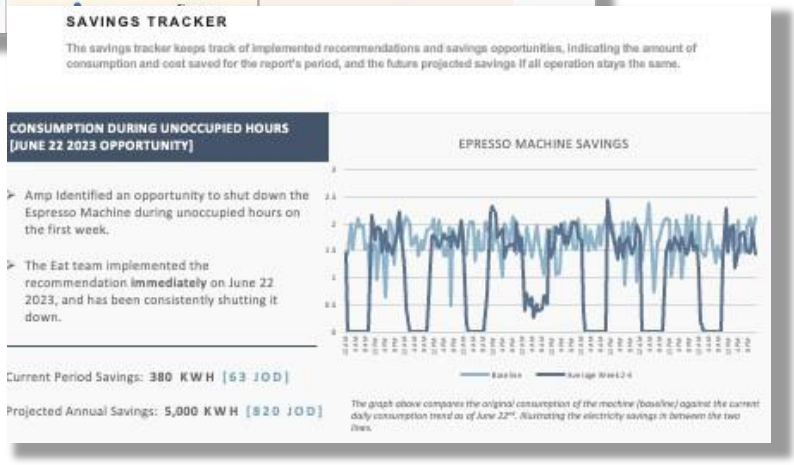


Key Features

Unlock Insights, Drive Change

Weekly/Monthly Energy Report





Summary report of performance delivered right to your fingertips

- Weekly and Monthly summary PDF reports delivered to users
- Email and platform access to acommodate user preferences
- High level data overview of performance per facility or building with equipment level data
- Shareable format to optimize stakeholder engagement and drive action
- Friendly data visualization to illustrate data including EUI, Equipment performance, Alerts recommendations and more.
- Standardized KPIs to track performance on weekly and monthly basis – seemless progress tracking

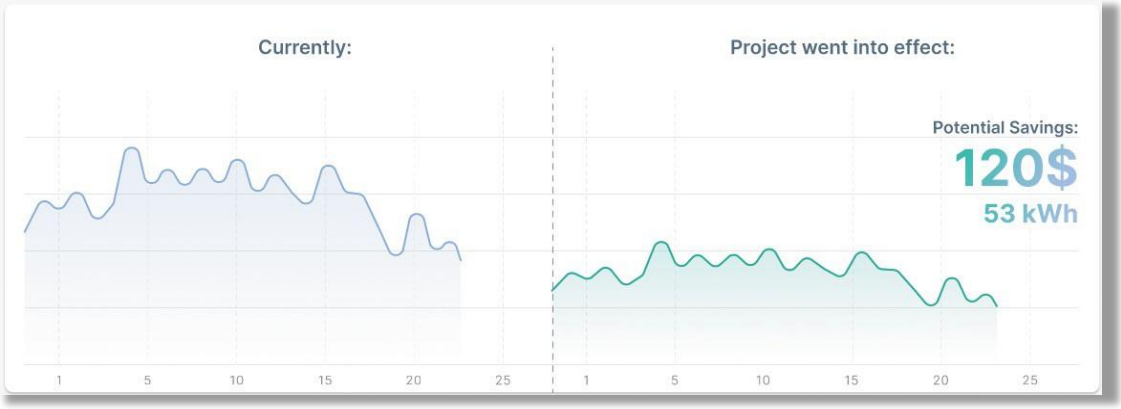
Key Features

Unlock Insights, Drive Change

AI-Powered Energy Manager

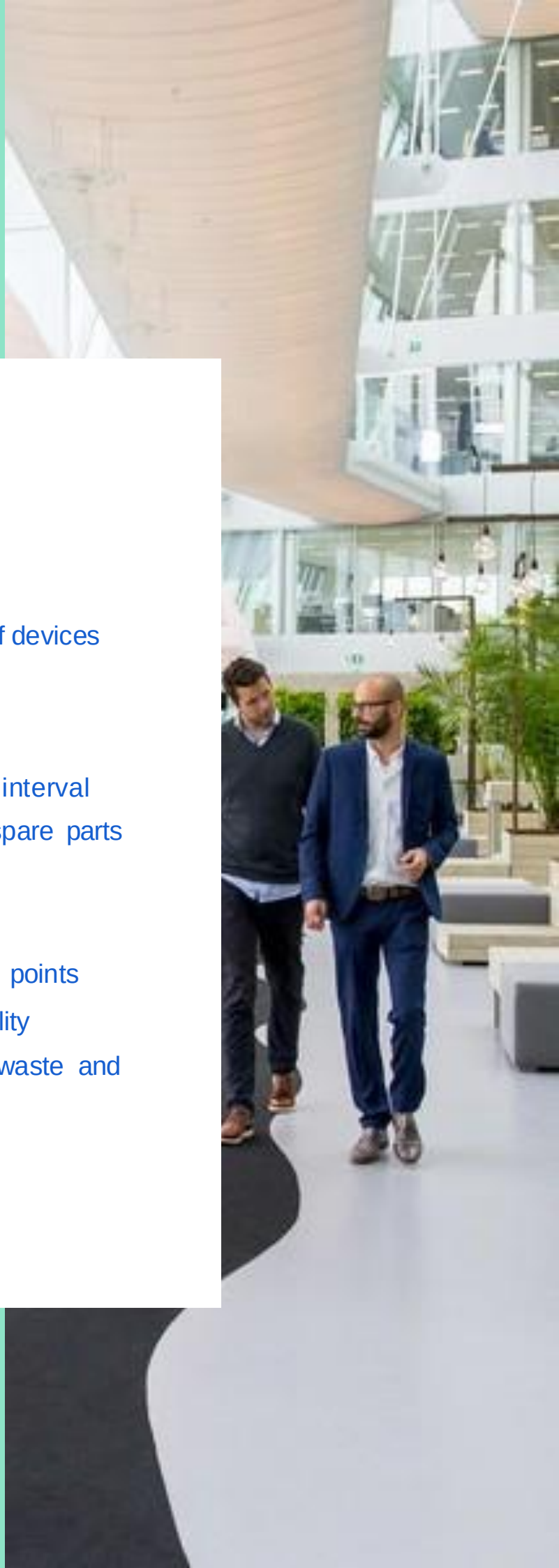
Get intelligent, real-time energy efficiency recommendations via our Energy Manager module:

- Tailored AI recommendations based on equipment-level data
- Actionable, real-time recommendations for on-ground implementation
- Visual baseline comparison of current versus optimal performance
- Displayed projected kWh and financial savings from enforcing recommendations
- Boost operational efficiency and reduce costs while implementing best practices
- Offers continuous commissioning, updating insights with real-time data



Engagement Details

- Fully Warranty on performance of devices and software during duration of subscription
- Default 1-minute data sample interval
- Full approvals, integration & spare parts services included
- Unlimited users
- Exploration of alternative data points including occupancy & air quality
- Clear identification of energy waste and inefficiencies



Contact Us

Learn More

Digital transformation can seem daunting, so we like to let our platform speak for itself. Reach out to us and schedule a Demo today. Let's talk about how IOT can seamlessly integrate into your organization.

hello@ampenergy.io

Floor 8, Tower 4, DWTC
Dubai, UAE

AMP

Think Smarter
Get Greener
Save Better